

Shubham Shinde

Research software and machine learning engineer · shubhamshinde.github.io

Bristol, UK shubham.edu.ms@gmail.com +44 7352 671872 github.com/ShubhamX57 linkedin.com/in/shubhams6

I build scientific and machine learning software, and I test whether its results can be trusted. MSc in Scientific Computing with Data Science from the University of Bristol, with four years of professional experience across a defence research programme, a science and innovation centre, a machine learning company and an engineering firm. Recent work spans a silent numerical failure in the *kinisi* diffusion library, a Bayesian neutron scattering library, a distributed Higgs boson analysis, and retrieval systems that refuse to report claims they cannot verify.

Education

2025 to 2026 Bristol, UK	MSc Scientific Computing with Data Science University of Bristol · Distinction expected · SARTHI Maharashtra Government Foreign Scholarship
2022 to 2023 Pune, India	PG Diploma in Data Science and Artificial Intelligence Savitribai Phule Pune University · GPA 8.5/10 · Thesis on building shadow detection from aerial imagery
2017 to 2020 Pune, India	BE Mechanical Engineering Savitribai Phule Pune University · CFD capstone on wind turbine blades · Merit Scholarship, 2018

Technical skills

Scientific computing Python, NumPy, SciPy, HDF5 and h5py, MDAnalysis, emcee, Matplotlib	Numerical methods Bayesian MCMC, numerical linear algebra, Monte Carlo methods, uncertainty quantification, model validation	High-performance computing SLURM, MPI, OpenMP, Cython, Numba, PySpark, Dask, multi-GPU; Monte Carlo code 4.7× faster than its NumPy version
Machine learning PyTorch, scikit-learn, XGBoost, LightGBM, OpenCV, MLflow, DVC, RAG, cross-encoder reranking, vector search	Engineering practice FastAPI, PostgreSQL, Docker, Kubernetes, RabbitMQ, Prometheus, GitHub Actions CI/CD, pytest, mypy, Git	Languages Python, SQL, Bash, JavaScript and TypeScript, MATLAB

Professional experience

Jun 2024 to Aug 2025 Pune, India	Data scientist and project engineer ARDE research programme, Raj Security and Facility Management Pvt Ltd - Designed authenticated, versioned FastAPI microservices serving sensor data to dashboards at sub-100 ms on a restricted offline network; Docker CI/CD cut release time from hours to under ten minutes. - Built a parallelised PySpark and SLURM pipeline ingesting over 20 GB of sensor data a day, with automated failure recovery, schema validation and audit logging into PostgreSQL for full traceability. - Developed deep learning models for elevation-angle prediction from sensor and imaging data, and an XGBoost and LightGBM ensemble that reduced prediction error by roughly 42% against the legacy analytical baseline. - Gathered requirements from stakeholder teams, reported results to engineers and management, and ran on-call production support with root-cause analysis, at a mean time to resolution under four hours.
Jun 2023 to May 2024 Satara, India	Associate fellow in robotics Rayat Science and Innovation Activity Centre - Developed AI features for educational robotics projects, integrating machine learning into robotic systems. - Trained object-recognition models on robot sensor data, implemented Python path-planning and navigation algorithms, and demonstrated the systems to researchers and students.
Feb 2023 to May 2023 Remote	Machine learning research engineer Spartifical Pvt Ltd Developed a hybrid CNN and Transformer classifier reaching 89.7% accuracy on NASA lunar terrain imagery, integrating NASA dataset APIs with internal annotation systems.

- Built a full-stack data service with FastAPI inference endpoints and a React monitoring dashboard, with pytest and Jest suites at **over 90%** coverage, packaged with Docker, DVC and CI for one-command reproduction.

Nov 2020 to May
2022
Satara, India

Design engineer

Gurudatta Enterprises Pvt Ltd

- Built 3D models and drawings in SolidWorks and CATIA V5 with tolerance stack-up analysis across **200+** prototype test cycles, and automated test-bench data logging in Python, cutting post-processing time by **roughly 60%**.

Research and open-source projects

Numerical stability of covariance matrix inversion in kinisi

2025 to 2026

MSc dissertation, University of Bristol, supervised by Dr Andrew McCluskey

- kinisi carries the full MSD covariance through a generalised least-squares fit; in some runs the matrix becomes non-positive-definite, silently degrading fitted coefficients and uncertainties. Built a random-walk testbed with known ground truth to separate artefact from signal.
 - Showed the failure rate fell from **90% to 2%** as sampling increased, isolating the cause to noise in long lag-time variance estimates; benchmarked eigenvalue, ridge and shrinkage preconditioning and proposed an adaptive eigenvalue floor that beat the fixed default.
- o Validation across noise levels is ongoing; a consistency check is open upstream as pull request #222.

gens: quasi-elastic neutron scattering analysis library

2025 to present

Author, open source · gens.readthedocs.io

- End-to-end toolbox for ISIS Mantid .nxspe data, combining Fickian, Chudley–Elliott and Singwi–Sjölander translational models and rotational structure factors in a forward model with measured-kernel resolution; validated on benzene data from LET.
- Joint likelihood across all Q-bins, multi-start MAP and emcee MCMC with automatic Gelman–Rubin diagnostics, and a plug-in model registry.

Distributed ATLAS H → ZZ* → 4ℓ Higgs analysis

2026

Open source · github.com/ShubhamX57/HZZ-Distributed-Computing

- Reproduces the ATLAS Higgs analysis at **4.5σ**, distributing **40** ROOT files across horizontally scaled Docker workers through a RabbitMQ coordinator–worker design.
- Exponential-backoff retries, a dead-letter queue, coordinator checkpointing for crash recovery, and Prometheus metrics.

arxiv-rag: grounded retrieval over research papers

2026

Open source · github.com/ShubhamX57/arxiv-rag

- Hybrid BM25 and BGE-M3 retrieval over **204** ML papers, ablated across six configurations; cross-encoder reranking lifted MRR@10 from **0.801 to 0.955**. **224** tests, strict mpy, CI.
- Found it attaching genuine citations to unsupported claims; built verbatim-quote verification with forced abstention, giving zero false grounding on out-of-corpus questions. Negative HyDE and multi-query results are published.

Physics-informed machine learning for pore characterisation

2026

Open source · github.com/ShubhamX57/biscuit-dunking-analysis

- Tested the Washburn capillary-flow equation on **3,000** measurements, then inverted it into a pore-radius feature: a random forest rose from **81% to 96%** accuracy, and a hybrid physics and ML pipeline lifted R² from **0.79 to 0.985**.

scisolve: agentic scientific computing with grounded answers

2026

Open source, MIT · github.com/ShubhamX57/scisolve

- Solves ODE, optimisation, fitting and linear algebra problems by executing NumPy and SciPy code; every reported number must trace to a computed value, and answers are checked by one of six independent verification methods. **122** tests on Python 3.10 to 3.13.
- o In development: five of seven milestones complete, tested through a scripted harness and not yet run against a live model API.

Publications and talks

When a covariance matrix silently breaks your diffusion analysis

Shinde, S. Lightning talk, RSECon26, UK Research Software Engineering Conference, Sheffield, September 2026.

Building shadow detection using aerial imagery with attention-enhanced YOLOv4

Shinde, S., et al. (2023). JETIR, 10(9). **Best Research Paper Award**, Savitribai Phule Pune University.